Tankless Water Heaters

MARY JAMES

Ed Wyatt is an engineering consultant and an assistant editor of *Home Energy*.

Q: It seems like I waste a lot of water while waiting for the hot water to get to my faucet. I have heard that tankless water heaters can provide instant hot water. How true is this, and can you recommend them?

—Impatient in Ohio

A: Tankless water heaters (also called demand or instantaneous water heaters) are designed to heat water as needed rather than draw it from a tank that stores preheated water. However, a tankless water heater is not an instant water heater. It can not cut the time it takes for the hot water to travel to your faucet. But the small size of many tankless water heaters allows them to be installed almost anywhere, which can translate into less waiting.

In general, tankless water heaters are both water and energy efficient, as they don't store hot water, but heat cold water only at the time of demand. Opening the hot water tap allows cold water to flow into the heater, and this flow triggers the heating element or burner to turn on. A valve or thermostat controls the heater to maintain the flowing water at a constant temperature. Instantaneous water heaters are available in propane, natural gas, or electric models.

In contrast, in a conventional tank—or storage—water heater, the heating element or burner periodically turns on throughout the day and night to heat water standing in a tank. The heating element turns on both to satisfy demand for hot water, but also to compensate for heat losses through the tank walls and up the flue (for gas heaters). Although newer, more energy-efficient storage models have been designed to reduce these standby losses, higher energy use by conventional water heaters has increased domestic interest in tankless water heaters.

Energy Factor (EF) is a good measure of the efficiency of a water heater. Although its technical definition has always lacked universal agreement, its value is determined using three criteria: recovery efficiency (which indicates how efficiently the heat from the energy source is transferred to the water); standby losses (which are defined as the percentage of heat lost per hour from the stored water compared to the heat content of the water); and cycling losses. The EF is determined through a Department of Energy (DOE) standard test procedure that involves a water heater that is put through a 24 hour simulated use test under control conditions. Increasing EF indicates increasing water heater efficiency.

EFs for gas tankless water heaters range from around 0.69 to 0.84, whereas EFs of standard gas storage heaters are between 0.5 and 0.65; the Energy Star minimum is 0.53. For electric tankless water heaters, the EF tends to be very close to 0.98. Conventional electric resistance water heaters generally have an EF between 0.75 and 0.95; some utility rebates establish a minimum EF for a 50-gallon tank of 0.88. If product literature from manufacturers doesn't give the appliance an EF rating, you can obtain it by contacting an appliance manufacturer association. At this time, no federal standards apply to energy factor requirements for tankless water heaters.

Go With the Flow

Tankless water heaters do have limitations. They are often touted as supplying "endless" hot water, and this is true to a point. Hot water will keep flowing as long as the rate of flow doesn't exceed the heater's ability to heat the water. This flow rate depends on the amount of temperature rise desired. The first tankless water heaters were developed in association with a particular use or appliance—for example, a shower or kitchen sink. The temperature control, that is, the flow valve, was not designed for multiple simultaneous uses. In practical terms, these systems were not meant for whole-house use. Most tankless heaters on the market are based on this approach. Because of this limitation on simultaneous use, the availability of hot water is indeed limited. If the difference between the supply water temperature and the desired water temperature is lowered, the capability of the water heater will increase; if the difference increases, the capability of the heater will decline.

A flow valve regulates water flow so that water is delivered at the selected temperature. As a result, tankless water heaters are rated by the number of gallons supplied per minute for a given temperature rise (from the cold water base temperature). Clearly, demand heaters are specified with both a maximum flow rate (usually 6-6.5 gpm) as well as a minimum flow rate for activation. Most demand heater switches require at least $\frac{3}{4}$ gpm to actually turn on. In addition, heater specifications establish a minimum and maximum water pressure under which they will operate (typically 15 to 150 psi).

Thus some large heaters may have minimum flow rates higher than desired for some uses; it may not be possible to get a trickle or splash of hot water. Although these units also can be used as boosters for solar water heating systems, for hot tubs, or for hydronic floor heating systems, this issue must be addressed; solar or hydronic systems may not require a minimum flow rate to turn on.

A tankless water heater must be properly sized in order to provide the maximum amount of hot water needed to supply the peak demand. This is important, because if the unit is of insufficient capacity, its users will be disappointed.

Each appliance that will be serviced by the demand heater and used at the same time must be accounted for, and its flow rate estimated. The sum of these rates will determine the peak demand. For faucets, assume 0.75-2.5 gallons (2.8-9.5 liters) per minute. For low-flow showerheads, figure 1.2-2 gallons (4.5-7.6 liters) per minute; older showerheads use approximately twice this rate. And clothes washers and dishwashers consume about 1-2 gallons (3.8-7.6 liters) per minute. The sum of these rates is the desired flow rate under peak demand conditions. Then, determine the input (cold water) temperature, and subtract it from the desired hot temperature (usually 120°F for all uses except dishwashing). Then the flow rate at the desired temperature rise can be compared to manufacturers' specifications.

Typically, whole-house demand heaters provide hot water at a rate of 2 to 4 gallons (7.6 to 15.2 liters) per minute. This flow rate might suffice if your household does not use hot water at more than one location at the same time (such as showering and doing laundry simultaneously); however, multiple concurrent uses can result in inadequate performance. If the flow rate (household demand for hot water) is maintained above the maximum rate of heating, then the water heater cannot catch up.

Costs

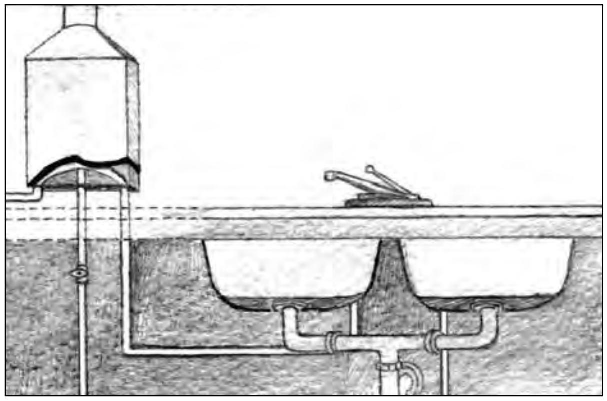
The cost of a tankless water heater varies from region to region. Small point-of-use heaters that deliver 1–2 gpm sell for about $200. Larger gas-fired tankless units that deliver 3–5 gpm can usually be found for $550–$1,000—although a few of these may be more costly and prices typically lie at the upper end of this range. To meet hot water demand when multiple faucets are being used, smaller demand heaters can be installed in parallel sequence.

In addition to unit costs, there are unique installation requirements. In order for tankless heaters to provide desired flow rates, they have much higher power requirements than typical water heaters do. For example, a conventional gas storage water heater might have a gas input of 40,000 BTU per hour; a gas tankless heater might need four times that power. A large tankless heater may have a maximum heat input of 600,000 BTU per hour. As a result, larger gas lines and vents may be needed.

For an electric unit, electrical service rated higher than what has been common in American homes must be available. This means that 100–125 amp circuitries will likely be insufficient, and that 150–200 amp (or more) services may be needed.

Because tankless water heaters depend on electronic control, both plumbing and electrical work must be done. It is possible to do the installation as a do-it-yourself project, but most manufacturers recommend professional installation. To avoid excessive redoing of piping and to minimize pumping costs, units are best placed near the points of use.

For gas heaters, consult National Fuel Gas Code NFPA 54. Indoor applications will require appropriate



vent piping for combustion exhaust. Each unit will provide minimum clearance around the heater for safety and operational concerns.

Some of the safety features that might be found in the unit, depending on its features, include an overheating sensor, a thermal fuse, a fan speed check, and/or a high current fuse. The heater must contain a safety relief valve activated at 150°F that complies with the Standard for Relief Valves and Automatic Shutoff Devices for Hot Water Supply Systems ANSI Z21.22.

Another cost consideration is the price of floor space. When new construction costs $200/ft^2, a conventional water heater might minimally occupy 4 ft^2 of floor area—more, if it needs its own utility closet. Imagine the cost savings with a wall-mounted demand heater that is installed in a laundry room, say, high enough for a washing machine to sit directly underneath it!

Payback times for tankless heaters under low use conditions are often longer than payback times for conventional heaters and longer than payback times for tankless heaters in high water use situations. Thus the most cost-effective situation for tankless heaters is in high flow rate applications. A fine line exists, however, given tank capacity and water temperature conditions; there is a maximum flow rate that cannot be exceeded. Thus the ideal unit must be chosen carefully.

Green Issues

Energy. Reduction in energy use is the principal environmental reason cited for encouraging greater use of tankless water heaters. A tankless water heater with intermittent ignition completely eliminates standby losses (heat lost through the walls of a storage tank by conduction and radiation), plus heat losses up the flue of gas-fired heaters and can reduce the cost of water heating by 20%–30%.

Water. Depending on household size and usage, water savings can be significant. Several studies have suggested that an average home of 3 to 4 persons can waste approximately 25 to 30 gallons of water per day down the drain waiting for hot water to be delivered to the tap—or upwards of 10,000 gallons of water per year. A tankless water heater, depending on its location within the home, can easily save 75%–90% of this total.

Solid waste. Tankless water heaters have an expected life span of 25 to 30 years—nearly twice that of tank heaters—for two basic reasons. First, these heaters are made with stainless steel and aluminum, with a solid copper or brass heat exchanger. Even the burners are stainless. And second, since water isn't stored in these devices, scale build-up is minimal. Most, (if not all) parts in tankless heaters are replaceable, extending the lives of these units—although some of these parts are not recyclable either.

Because demand water heaters last longer than, and are smaller than, conventional tanks, they use fewer materials. The long-term material consumption of tankless water heater use is significantly lower than that of conventional water heating systems.

A tankless water heater is one of the more efficient water heating solutions you can install, saving energy and water, and providing comfortable water quickly. The key to choosing this appliance is to understand the maximum flow rates for your use.